POTPRESS · USER REQUIREMENTS

PotPress user requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-25	Amish Chadha	PR-013 decided by Amish (D8, 2026-10-25)
0.2	2026-10-24	Amish Chadha	UR-013 now traced to PR-013 (kit 1.9.1 upgrade); no user requirement changed
0.1	2026-10-23	Amish Chadha	First user requirements, drawn from PPR-PRB-001 Tables 1 and 2 (kit 1.8 upgrade)

PotPress user requirements

What filter factories, fabrication shops, QC technicians and labs need, written from their side. They are drawn from the users and stakeholders in PPR-PRB-001 and are not yet validated with a filter factory; co-design sessions will revise them. Product requirements that meet these needs are in PPR-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix PPR-RTM-001 (`03b-traceability.md`). Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Standard filter	Form the factory's standard flowerpot filter, about 10 L working volume	High	Small factory (PRB Table 1)
UR-002	Performance	Even walls	Walls even enough, pot to pot, that fired filters flow inside the factory's band	High	Small factory; QC technician
UR-003	Performance	Daily output	Press 50 or more filters a day with two workers	High	Small factory
UR-004	Deployment and assembly	Local build and repair	Build and repair the frame with a stick welder, grinder, drill press and hand tools	High	Fabrication shop; NGO
UR-005	Deployment and assembly	Local molds	Have the molds cast locally from open files	Medium	Fabrication shop
UR-006	Logistics	Two-person handling	Move and assemble every part with two people and simple lifting gear	Medium	Fabrication shop; small factory
UR-007	Safety	No reach into the press	Operators cannot reach a pinch point while force is applied	High	Small factory; safety authority (PRB Table 2)
UR-008	Usability	Same flow test every time	Test every fired filter for flow and record it the same way each time	High	QC technician
UR-009	Cost	Affordable	Cost well below an imported press, within a start-up budget	High	NGO; small factory
UR-010	Environment and durability	Fits the workshop	Fit an existing workshop floor without mains power	Low	Small factory
UR-011	Safety	Safe for drinking water	Nothing in the press or test rig contaminates the clay or the test water	High	Households; NGO (PRB Table 2)
UR-012	Performance	Repeatable settings	Know and repeat the pressing setting between batches	Low	University or research lab
UR-013	Maintenance	Local spares	Replace wear parts from regional markets	Medium	Small factory

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.

UR-013 (local spares) is met by PR-013, added on 2026-10-24 and decided by Amish on 2026-10-25 (D8).